L62 – NOTICE BOARD CLI PROJECT REPORT

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Project Title: Notice Board CLI Using Bash Shell Scripting
Platform: Ubuntu Linux / WSL

1. Introduction

The Notice Board CLI is a terminal-based application developed using Bash shell scripting. It allows users to manage college notices through a simple command-line interface.

The project demonstrates Linux commands, shell scripting, file handling, searching, sorting, backup operations and task automation.

2. Objectives

- To develop a notice management system using Bash.
- To add and display college notices.
- To search notices using keywords.
- To identify HIGH priority notices.
- To archive notices older than 30 days.
- To create backups of notice data.
- To automate archiving using cron.

3. Technologies Used

- Operating System: Ubuntu Linux
- Environment: Windows Subsystem for Linux (WSL)
- Language: Bash Shell Scripting
- Commands: grep, sort, head, tar, date, mkdir, chmod
- Scheduler: Cron
- Editor: Nano

4. Project Structure

- main.sh – Main application and menu.
- lib/functions.sh – Functions used by the application.
- data/notices.txt – Stores notice records.
- data/archive.txt – Stores archived notices.
- backups/ – Stores backup files.
- logs/activity.log – Stores activity records.
- reports/ – Stores project reports and cron output.

5. Features

1. Post a new notice with the current date.
2. Display the latest five notices.
3. Search notices using keywords.
4. Display HIGH priority notices.
5. Archive notices older than 30 days.
6. View all notice data.
7. Create compressed backups using tar.
8. View activity logs.

6. Automation Using Cron

A cron job is configured to execute the archive function every Monday at 8:00 AM.

Cron expression:

0 8 * * 1

Command:

```
cd /home/ravi/linux_project/noticeboard &&
/home/ravi/linux_project/noticeboard/main.sh --cron
```

The cron service was checked and confirmed as active. The cron execution log was also verified.

7. Testing and Results

The following operations were tested:

- Help command.
- Backup command.
- Cron command.
- Main application menu.
- Notice management features.
- Activity logging.

The notice data file contains 25 sample notices and one header row. The backup operation generated a compressed archive successfully.

8. Conclusion

The Notice Board CLI project successfully demonstrates the practical use of Bash scripting and Linux commands. It provides a simple way to manage notices, search information, archive old records, create backups and automate repetitive tasks using cron.

This project improved my understanding of Linux file management, shell scripting and task scheduling.